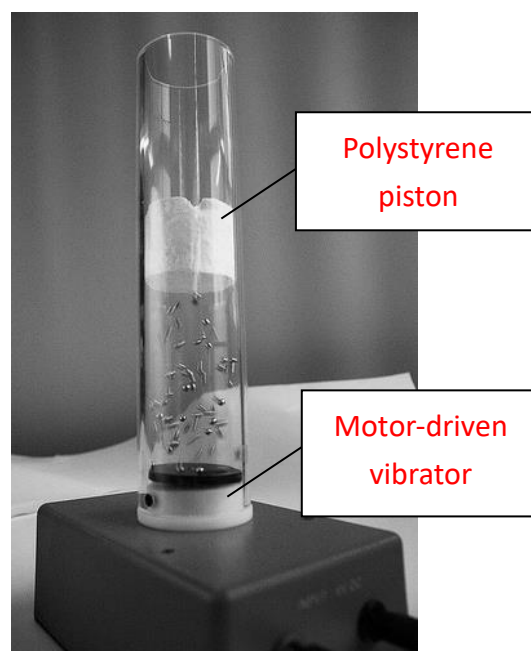


Kinetic theory model of an ideal gas

Scientist Brown finds that the air molecule is **in random motion**. The relationship between pressure, temperature and volume of the gas should be related to the behavior and motion of the air molecules. We find that **kinetic theory model** is very successful in explaining the gas law.

	Kinetic theory model
Air molecule N	Ball bearings
Pressure P	Sound produced when ball bearings hit the tube
Temperature T	Voltage applied to motor
Volume V	Height of piston



Assumption for Ideal Gas Law

- All molecules are identical and move randomly
- No intermolecular force between gas molecules
- Size of gas molecule is negligible compared to the volume of gas
- All collisions are elastic
- Impact time is short compare to the duration between collision

Explain how gases exert pressure on the wall of containers by kinetic theory model

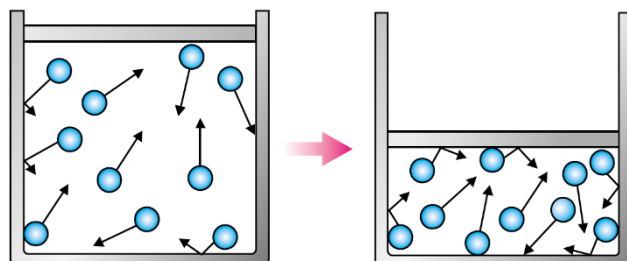
- The momentum of a gas molecule is changed when it collides with the wall of a container and then rebounds.
- By Newton's second law of motion, there must be a net force acting on the gas molecule by the wall.
- By Newton's third law of motion, the molecule must exert a force of same magnitude on the wall in the opposite direction and pressure is resulted finally.
- The pressure is measured by rate of change of momentum of the gas molecules per unit surface area.

(i) Using Kinetic theory to explain Boyle's law

- When the gas is compressed at constant temperature, the average speed of the gas molecules remains unchanged.
- As the volume decreases, gas molecules collide the walls more frequently, thus pressure increases.

Remarks 1: As temperature remains constant, kinetic energy, speed and momentum of molecules are unchanged.

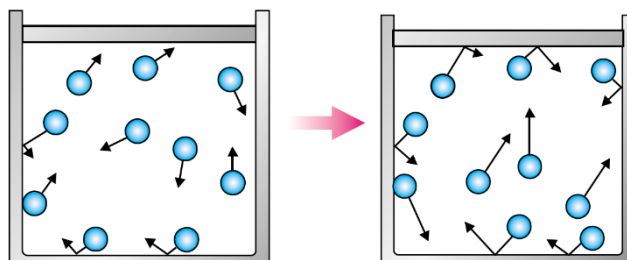
Remarks 2: As volume is decreased, the gas separation is decreased and the density is increased.



(ii) Using Kinetic theory to explain Pressure-Temperature relation

- As temperature increases, average kinetic energy of molecules increases and molecules move faster.
- Gas molecules bombard the wall of container more violently and frequently. Thus, pressure increase.

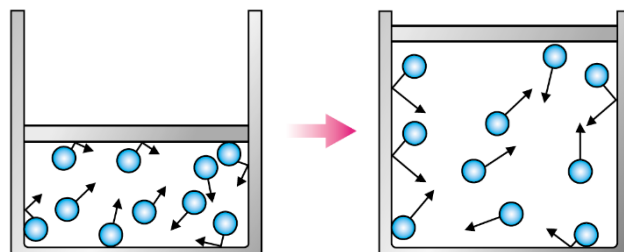
Remarks : As volume is constant, gas separation and gas density remains unchanged.



(iii) Using Kinetic theory to explain Volume-Temperature relation

- When temperature increases, the kinetic energy of gas molecules increases and thus the molecules move faster.
- Since the molecules bombard the wall more violently, the volume must increase so that the molecules then bombard the wall less frequently to give same pressure.

Remarks : As volume is increased, the gas separation increases and the gas density decreases.



Kinetic theory equation of an ideal gas

- Suppose a container with volume V contains N gas molecules, each of mass m . Due to the collision of gas molecules onto the walls of the container, the gas exerts a pressure P .

$$PV = \frac{1}{3} Nm \overline{c^2}$$

Where P is gas pressure exerting on the walls of a container

V is volume of the container

N is the number of molecules

m is the mass of a gas molecules

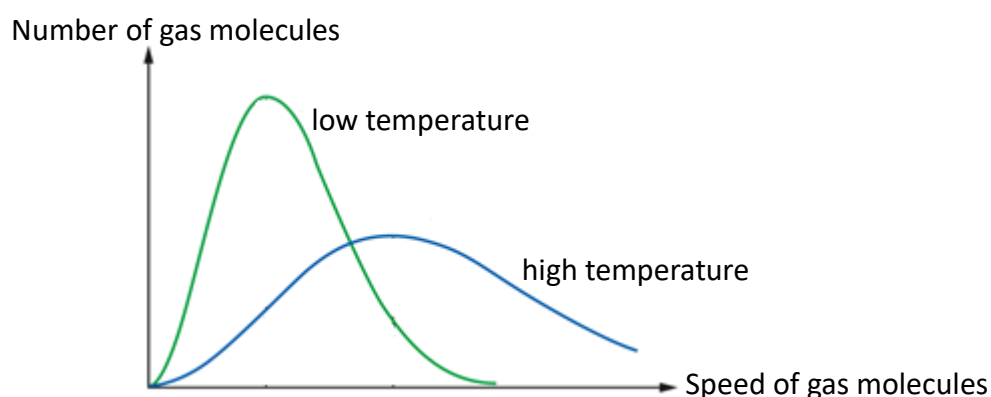
$\overline{c^2}$ is the **mean square speed** of the molecules

- Mean square speed of gas molecules $\overline{c^2}$ is calculated by taking average of the velocities of all the gas molecules in the container:

$$\overline{c^2} = \frac{c_1^2 + c_2^2 + \dots + c_N^2}{N}$$

($c_1, c_2, c_3, \dots, c_N$ = speed of N gas molecules)

- Root-mean-square speed ($\sqrt{\overline{c^2}}$ or v_{rms}) is the **root** of the mean of the square speed of molecules. Root-mean-square speed is not equal to average speed of gas molecules.
- Gas molecules at a certain temperature moves with different speeds. The distribution depends on the **temperature**.



Root mean square speed of gas molecules

Combine two equations for an ideal gas:

$$PV = \frac{1}{3} Nm \overline{c^2} \quad \text{and} \quad PV = nRT$$

$$nRT = \frac{1}{3} Nm \overline{c^2}$$

$$RT = \frac{M \overline{c^2}}{3}$$

$$\overline{c^2} = \frac{3RT}{M}$$

$$V_{rms} = \sqrt{\overline{c^2}} = \sqrt{\frac{3RT}{M}}$$

Remarks:

$$n = \frac{Nm}{M}$$

Where n is number of moles of gas molecules
 N is the number of molecules
 m is the mass of a gas molecules
 M is the **molar mass** of the gas

Relation with the density ρ of the gas:

$$PV = \frac{1}{3} Nm \overline{c^2}$$

$$P = \frac{1}{3} \frac{Nm}{V} \overline{c^2}$$

$$P = \frac{1}{3} \rho \overline{c^2}$$

$$\overline{c^2} = \frac{3P}{\rho}$$

$$V_{rms} = \sqrt{\overline{c^2}} = \sqrt{\frac{3P}{\rho}}$$

Remarks:

$$\rho = \frac{Nm}{V}$$

Where N is the number of molecules
 m is the mass of a gas molecules
 ρ is the density of the gas

Average kinetic energy of a molecule

Combine two equations for an ideal gas:

$$PV = \frac{1}{3} N m \overline{c^2} \quad \text{and} \quad PV = nRT$$

$$nRT = \frac{1}{3} N m \overline{c^2}$$

$$\frac{N}{N_A} RT = \frac{1}{3} N m \overline{c^2}$$

$$\frac{1}{N_A} RT = \frac{1}{3} m \overline{c^2}$$

$$KE_{\text{average}} = \frac{1}{2} m \overline{c^2} = \frac{3}{2} \frac{R}{N_A} T$$

Remarks:

Where N_A is the Avogadro's constant (6.02×10^{23})

n is number of moles of gas molecules

N is the number of molecules

m is the mass of a gas molecules

$$KE_{\text{average}} \propto T \text{ (in kelvin)}$$

Internal energy of an ideal gas

Since the **inter-molecular forces** in an ideal gas are **negligible**, ideal gas has **no molecular potential energy**. Internal energy of an ideal gas is then equal to **total kinetic energy** of the gas molecules.

For total N molecules, its kinetic energy:

$$KE_{\text{total}} = N \times \frac{3}{2} \frac{R}{N_A} T$$

$$KE_{\text{total}} = \frac{3}{2} nRT$$

Real gas Vs Ideal gas

In the assumption for ideal gas, there are 2 assumptions that real gas cannot be obeyed:

1. The intermolecular attraction forces are negligible for an ideal gas.
2. The size of a gas molecule is very small and negligible for an ideal gas.

For real gas:

1. The intermolecular attraction forces exist between molecules
2. The gas molecules have sizes which is not negligible

Remarks: real gas approaches ideal gas behaviour only at

- Low pressure
- High temperature

1. Which of the following behaviors can be simulated from this kinetic theory model of gas ?

- (1) Increasing temperature at constant volume causes pressure to increase.
- (2) Increasing volume at constant pressure causes temperature to increase.
- (3) Increasing temperature at constant pressure cause volume to increase.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

- (1) ✓ If the power of motor increases (T increases), more weight has to be added (p increases) so as to keep the height of the column constant (V constant).
- (2) ✗ There is no direct way to 'increase the volume' under constant pressure. Even we change the height of column (volume), the power of motor remains unchanged.
- (3) ✓ If the power of motor increases (T increases), height of the column increases (V increases) and there is no change in weight (p constant).

B

2. When gas molecules strike the wall of the container, which of the following statements are incorrect ?

- (1) The gas molecules move faster.
- (2) The gas molecules lose kinetic energy.
- (3) The wall of the container exerts a force on the gas molecules.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

- (1) ✗ The kinetic energy of the gas molecules is conserved when they hit the wall of the container as they collide elastically. So they move in same speed, but in opposite direction.
- (2) ✗ See (1).
- (3) ✓ When gas molecules hit the wall, they exert a force to the wall, giving rise to pressure. By Newton's 3rd Law, the wall also exerts a force of equal magnitude (but opposite in direction) to the gas molecules.

A

3. A gas is stored in a container. Which of the following changes will increase the frequency of collisions between the gas molecules and the wall of the container?

- (1) Increase the volume of the container.
- (2) Increase the number of gas molecules inside the container.
- (3) Increase the temperature of the gas.

- A. (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. All of the above

C Volume increases

⇒ a gas molecule takes a longer time to move from one wall to another

⇒ frequency of collisions decreases

∴ (1) is incorrect.

Number of gas molecules increases

⇒ within the same period of time, more gas molecules collide with the wall

⇒ frequency of collisions increases

∴ (2) is correct.

Temperature increases

⇒ gas molecules move faster

⇒ a gas molecule takes a shorter time to move from one wall to another

⇒ frequency of collisions increases

∴ (3) is correct.

C

4. Which of the following is **not** the assumption of the kinetic theory of an ideal gas?

- A. All molecules are identical.
- B. All molecules are in random motion.
- C. The size of each molecules is negligible.
- D. The kinetic energy of every molecule is the same.

D

5. Which of the following is **not** the assumption of the kinetic theory of an ideal gas?

- A. All molecules are in random motion.
- B. All molecules move with same speed at a certain temperature.
- C. All molecules are point particles that have no physical size.
- D. All collisions are perfectly elastic.

B

- ✗ A. Molecules of ideal gas are assumed to move in random motion.
- ✓ B. Speed of molecules should have a distribution, with an average value of speed.
- ✗ C. Molecules of ideal gas are assumed to have no volume.
- ✗ D. Molecules of ideal gas are assumed to have perfectly elastic collision.

6. In kinetic theory equation $pV = \frac{1}{3} Nmc^2$, what does the symbol c^2 mean?

- A. root mean square speed of gas
- B. mean square speed of gas
- C. mean speed of gas
- D. speed of light

B

7. In kinetic theory equation $pV = \frac{1}{3} Nmc^2$, what does the symbol c mean?

- A. root mean square speed of gas
- B. mean square speed of gas
- C. speed of gas
- D. speed of light

Note:

Speed of gas (c) $\neq$ root mean square speed

($c_{rms} = \sqrt{c^2}$) $\neq$ mean speed ($\bar{c}$)

Mean square speed ($\overline{c^2}$) $\neq$ square of mean speed ($\bar{c}^2$)

C

8. HKAL 2008 MC Q.22

Which of the following assumptions leads directly to the factor $\frac{1}{3}$ in deriving the equation

$pV = \frac{1}{3} Nmc^2$ in kinetic theory of gases?

- A. All gas molecules are in random motion.
- B. The gas molecules are negligible size.
- C. All collisions are perfectly elastic.
- D. There is no intermolecular force between molecules except during collision.

Then we assume that the particle is in random motion,
 $\therefore v_x^2 = v_y^2 = v_z^2$
 $\therefore \overline{c^2} = \frac{1}{3} \overline{v_x^2}$
 So there is a factor $\frac{1}{3}$.

A

9. The mass and temperature of a gas are 100 g and 210 °C respectively. The root-mean-square speed of the gas molecules is 650 ms⁻¹. What is the number of moles of the gas?

- A. 0.12 mol
- B. 0.28 mol
- C. 3.5 mol
- D. 8.1 mol

C By $c_{rms} = \sqrt{\frac{3pV}{Nm}}$ and $pV = nRT$,

$$n = \frac{Nmc_{rms}^2}{3RT} = \frac{0.1 \times 650^2}{3 \times 8.31 \times (273 + 210)} = 3.5 \text{ mol}$$

C

10. Two ideal gases X and Y are at the same temperature. The molar mass of X and Y are 32 and 20 respectively. If the r.m.s speed of a gas molecule in X is v , find the r.m.s speed of a gas molecule in Y

- A. $\frac{\sqrt{5}}{\sqrt{8}} v$
- B. $\frac{\sqrt{8}}{\sqrt{5}} v$
- C. $\frac{5v}{8}$
- D. $\frac{8v}{5}$

$$\frac{1}{2} mv^2 = \frac{3}{2} kT$$

$$v \propto \frac{1}{\sqrt{m}}$$

$$\frac{v'}{v} = \sqrt{\frac{32}{20}}$$

$$v' = \sqrt{\frac{8}{5}} v$$

The answer is B.

B

11. The molecular mass of helium and neon are 4 and 20 respectively. If a bottle of helium gas has the same r.m.s speed as a bottle of neon gas, find the ratio of helium gas to the temperature of neon gas

- A. $1 : \sqrt{5}$
B. $1 : 5$
C. $\sqrt{5} : 1$
D. $5 : 1$

$$\frac{1}{2}mv^2 = \frac{3}{2}kT$$

$$T \propto m$$

$$T_{\text{helium}} : T_{\text{neon}} = m_{\text{helium}} : m_{\text{neon}} = 4 : 20 = 1 : 5$$

The answer is B.

B

12. Two gases X and Y are at the same temperature. The mass of one mole of gas X is 16 times that of Y. What is the ratio of the root-mean-square speed of the molecules of X to that of Y.

- A. $1 : 2$
B. $2 : 1$
C. $1 : 4$
D. $1 : 16$

C By $c_{\text{rms}} = \sqrt{\frac{3RT}{mN_A}}$, at constant temperature,
ratio of c_{rms} of the molecules of X to that of Y

$$= \frac{\sqrt{\frac{1}{m_X}}}{\sqrt{\frac{1}{m_Y}}} = \sqrt{\frac{m_Y}{m_X}} = \sqrt{\frac{1}{16}} = \frac{1}{4}$$

C

13. A gas is at 27°C . Find the kinetic energy of one molecule. (Avogadro number $N_A = 6.02 \times 10^{23}$ and universal gas constant $R = 8.31 \text{ Jmol}^{-1}\text{K}^{-1}$)

- A. $5.59 \times 10^{-22} \text{ J}$
B. $6.21 \times 10^{-21} \text{ J}$
C. 337 J
D. 7480 J

$$\text{Internal energy} = \frac{3RT}{2N_A} = \frac{3 \times 8.31 \times (273 + 27)}{2 \times 6.02 \times 10^{23}}$$

$$= 6.21 \times 10^{-21} \text{ J}$$

The answer is B.

B

14. In which of the following situations would the r.m.s speed of the molecules of a fixed mass of an ideal gas increase?

- (1) The gas is heated under constant volume.
(2) The gas expands under constant pressure.
(3) The gas is compressed under constant temperature

- A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

✓	(1)	As the gas is heated, its temperature must increase, thus r.m.s. speed would increase.
✓	(2)	At constant pressure, the gas expands only when temperature increases, thus r.m.s. speed must increase.
✗	(3)	Under constant temperature, average kinetic energy and r.m.s. speed remain unchanged.

C

15. Two vessels contain hydrogen and oxygen gas respectively. Both gases have the same pressure and temperature and are assumed to be ideal. Which of the following physical quantities must be the same for the two gases?

- A. The volume of the gas
B. The mass per unit volume of the gas
C. The r.m.s speed of the gas molecules
D. The number of gas molecules per unit volume

$$\text{By } PV = nRT$$

$$\therefore \frac{n}{V} = \frac{P}{RT}$$

Since P and T are the same, the ratio $\frac{n}{V}$ is the same for the two gases.

As the number of gas molecules N is proportional to the number of mole n , they have the same number of gas molecules per unit volume $\frac{N}{V}$.

D

16. (a) Explain what is meant by the root-mean-squared speed for an assembly of molecules.
(b) At a certain time the speeds of seven molecules are as shown.

Speed/ms ⁻¹	2.0	3.0	4.0	5.0	6.0
Number	1	3	1	1	1

- (i) Calculate the root-mean-squared speed for this group of molecules.
(ii) Find the average speed.

(a) For a group of N molecules with velocities $v_1, v_2, v_1, \dots, v_N$, the root-mean-squared speed is the square root of the average of the sum of the square of the speeds for all the molecules.

(b) (i) $c = \{1/7 \times (1 \times 2^2 + 3 \times 3^2 + 1 \times 4^2 + 1 \times 5^2 + 1 \times 6^2)\}^{1/2}$
 $= 3.9 \text{ ms}^{-1}$

(ii) Average speed = $(1 \times 2 + 3 \times 3 + 1 \times 4 + 1 \times 5 + 1 \times 6)$
 $= 3.7 \text{ ms}^{-1}$

17. The root-mean-square speed of the molecules of a certain gas at 400 K is 1000 ms^{-1}

- (a) Find the average kinetic energy of the molecules. (2 marks)
(b) Find the mass of one molecule of the gas. (2 marks)

(a) Average kinetic energy of the molecules = $\frac{3RT}{2N_A}$ 1M
 $= \frac{3 \times 8.31 \times 400}{2 \times 6.02 \times 10^{23}}$
 $= 8.28 \times 10^{-21} \text{ J}$ 1A

(b) By $\text{KE}_{\text{average}} = \frac{1}{2} m c_{\text{rms}}^2$, 1M
 $m = \frac{2 \times \text{KE}_{\text{average}}}{c_{\text{rms}}^2}$
 $= \frac{2 \times 8.28 \times 10^{-21}}{1000^2}$
 $= 1.66 \times 10^{-26} \text{ kg}$ 1A
 $\therefore$ The mass of one molecule of the gas is $1.66 \times 10^{-26} \text{ kg}$.

18. A container stores an ideal gas at 47 °C. The mass of 1 mole of the gas is 0.002 kg.

- (a) Find the average kinetic energy of the gas molecules. (2 marks)
(b) Find the root-mean-square speed of the gas molecules. (2 marks)
(c) How will the answers in (a) and (b) change if the temperature of the gas decreases?

(a) Average kinetic energy of the gas molecules = $\frac{3RT}{2N_A}$ 1M
 $= \frac{3 \times 8.31 \times (273 + 47)}{2 \times 6.02 \times 10^{23}}$
 $= 6.63 \times 10^{-21} \text{ J}$ 1A

(b) Root-mean-square speed of the gas molecules = $\sqrt{\frac{3RT}{mN_A}}$ 1M
 $= \sqrt{\frac{3 \times 8.31 \times (273 + 47)}{0.002}}$
 $= 2000 \text{ m s}^{-1}$ 1A

(c) Both answers will decrease. 1A + 1A

19. In a certain room, the temperature, pressure and density of the air are 25°C, 101.3 kPa and 1.2 kg m⁻³ respectively

(a) Find the root-mean-square speed of the air molecules.

(b) What is your answer in (a) if the temperature rises to 63°C and the pressure remains unchanged?

$$(a) \text{ By } p = \frac{m}{V} N \left(\frac{\overline{c^2}}{3} \right),$$

$$c = \sqrt{\frac{3p}{\rho}}$$

we have

$$c = \sqrt{\frac{3(101.3 \times 1000)}{1.2}} = 503.2 \text{ ms}^{-1}$$

(b) Since $\overline{c^2} \propto T$

$$\frac{\overline{c^2(63^\circ\text{C})}}{\overline{c^2(25^\circ\text{C})}} = \frac{T_{63^\circ\text{C}}}{T_{25^\circ\text{C}}}$$

$$\begin{aligned} \overline{c^2(63^\circ\text{C})} &= 503.2 \times \frac{\sqrt{63+273}}{\sqrt{25+273}} \\ &= 534 \text{ ms}^{-1} \end{aligned}$$

20. 2017 DSE Q.3

The average kinetic energy of one monatomic gas molecule at temperature T is given by

$$E_K = \frac{3}{2} \left(\frac{R}{N_A} \right) T,$$

Where R is the universal gas constant and N_A is the Avogadro constant. A monatomic gas is heated from 300 K to 350 K under fixed volume.

(a) Estimate the ratio of the root-mean-square speed ($c_{r.m.s.}$) of the gas molecules at the two temperatures ($\frac{c_{r.m.s. \text{ at } 350 \text{ K}}}{c_{r.m.s. \text{ at } 300 \text{ K}}}$) (2 marks)

(b) Thus, using kinetic theory, explain why the gas pressure would increase. (2 marks)

$$(a) \frac{(c_{rms})_f}{c_{rms}} = \sqrt{\frac{T_f}{T}}$$

$$\begin{aligned} \frac{c_{r.m.s. \text{ at } 350 \text{ K}}}{c_{r.m.s. \text{ at } 300 \text{ K}}} &= \sqrt{\frac{350}{300}} \\ &= 1.08 \end{aligned}$$

1M

1A

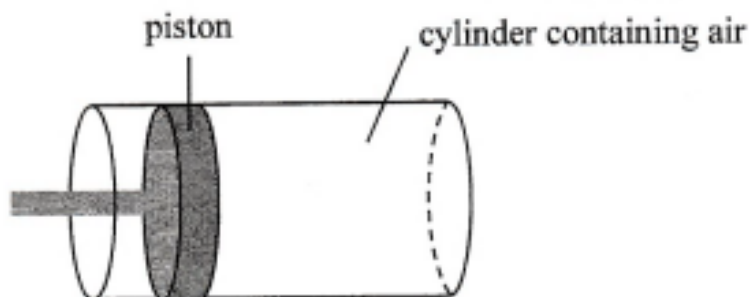
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(b) The speed of the gas molecules increases. They collide more frequently and violently with the wall of the container. Thus, the pressure increases.

1A+1A

2

21. A cylinder containing air is fitted with a piston which can move smoothly. The air inside the cylinder has a volume of 350 cm^3 and is at the atmospheric pressure of 102 kPa . The mass of air inside the cylinder is 0.42 g . The molar mass of air is 29 gmol^{-1} . Assume that air behaves like an ideal gas.
- (a) Calculate the Celsius temperature of air inside the cylinder. (2 marks)
- (b) Calculate the root-mean-square speed of the air molecules inside the cylinder. (2 marks)
- (c) The cylinder is now placed in a water bath containing boiling water. It is allowed to reach thermal equilibrium while keeping the pressure of air at atmospheric pressure. Based on the kinetic theory of gases, explain why the volume of air increases with temperature under constant pressure. (2 marks)



- (a) $PV = nRT$
 $\therefore (102 \times 10^3) \times (350 \times 10^{-6}) = \left(\frac{0.42}{29}\right) \times (8.31) \times (\theta + 273)$ [1]
 $\therefore \theta = 23.6^\circ\text{C}$ [1]
- (b) $PV = \frac{1}{3}Nm c^2 = \frac{1}{3}M c^2$
 $\therefore (102 \times 10^3) \times (350 \times 10^{-6}) = \frac{1}{3} \times (0.42 \times 10^{-3}) c^2$ [1]
 $\therefore c = 505 \text{ m s}^{-1}$ [1]
- OR**
- $P = \frac{1}{3} \rho c^2$
 $\therefore (102 \times 10^3) = \frac{1}{3} \times \left(\frac{0.42 \times 10^{-3}}{350 \times 10^{-6}}\right) \times c^2$ [1]
 $\therefore c = 505 \text{ m s}^{-1}$ [1]
- OR**
- $c = \sqrt{\frac{3RT}{M}}$
 $= \sqrt{\frac{3 \times (8.31) \times (23.6 + 273)}{(29 \times 10^{-3})}}$ [1]
 $= 505 \text{ m s}^{-1}$ [1]
- (c) When the temperature of air increases,
 air molecules move faster and hit the walls of container more violently. [1]
 To give same pressure,
 the air molecules hit the walls less frequently [1]
 by increasing the volume.

22. Two containers *A* and *B* with volumes 100 cm^3 and 500 cm^3 respectively are connected by a tube of negligible volume as shown in Figure 10.1. The tap *T* for controlling gas flow is closed initially. Container *A* contains an ideal gas at a pressure of $12 \times 10^5 \text{ Pa}$ while there is a vacuum in container *B*. The temperature of the two containers is maintained at 0°C by two separate water baths with melting ice.

(Given: Universal gas constant = $8.31 \text{ J K}^{-1} \text{ mol}^{-1}$)

Avogadro constant = $6.02 \times 10^{23} \text{ mol}^{-1}$

Mass of a molecule of the ideal gas = $4.52 \times 10^{-26} \text{ kg}$)

- Calculate the number of moles of gas in container *A*.
- When tap *T* is open and steady state is reached, find the root-mean-square speed of the gas molecules in *A*?
- The water bath for container *A* is heated until the water boils. The tap *T* remains open. Calculate
 - The equilibrium pressure of the gas in the two containers; and
 - The net amount of gas, in moles, that passed through the connecting tube during the heating process.

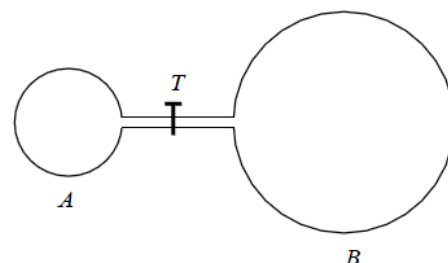


Figure 10.1

$$\begin{aligned} \text{(a)} \quad pV &= nRT \\ (12 \times 10^5) (100 \times 10^{-6}) &= n (8.31) (273) \\ n &= 0.0529 \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad pV &= \frac{1}{3} N m \overline{c^2} \\ (12 \times 10^5) (100 \times 10^{-6}) &= \frac{1}{3} (0.0529) (6.02 \times 10^{23}) (4.52 \times 10^{-26}) \overline{c^2} \\ \sqrt{\overline{c^2}} &= 500 \text{ m s}^{-1} \end{aligned}$$

$$\begin{aligned} \text{(i)} \quad n_1 + n_2 &= n_0 \\ \frac{pV_1}{RT_1} + \frac{pV_2}{RT_2} &= n_0 \\ \frac{p}{8.31} \left(\frac{100 \times 10^{-6}}{273} + \frac{500 \times 10^{-6}}{273} \right) &= 0.0529 \\ p &= 2.09 \times 10^5 \text{ Pa} \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad n_1 &= \frac{pV_1}{RT_1} = \frac{(2.09 \times 10^5) (100 \times 10^{-6})}{(8.31) (273)} \\ &= 0.0068 \text{ mole} \\ \text{Net amount of gas flow from } A \text{ to } B &: \frac{0.0529}{6} - 0.0068 = 0.002 \text{ (mole)} \end{aligned}$$

23. Suppose there are one mole of helium gas and one mole of neon gas at 27°C . The relative molecular masses of helium and neon are 4 and 20 respectively. Assume both gases behave as an ideal gas.

- Find the average kinetic energy of the helium gas molecules.
- Find the ratio of the following quantities of helium gas and neon gas.
 - average kinetic energy of molecules
 - root-mean-square speed of the molecules
- If the temperature of the helium gas is raised by 50°C at a constant volume, how much energy is required?

(Given that the universal gas constant $R = 8.31 \text{ J mol}^{-1}\text{K}^{-1}$ and Avogadro number $N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$)

(a) Applying $\text{K.E.}_{\text{avg}} = \frac{3}{2}kT$ and $k = \frac{R}{N_A}$, we have $\text{K.E.}_{\text{avg}} = \frac{3RT}{2N_A}$

$$\text{K.E.}_{\text{avg}} = \frac{3 \times 8.31 \times (27 + 273)}{2 \times (6.02 \times 10^{23})}$$

$$\approx 6.21 \times 10^{-21} \text{ J}$$

The average kinetic energy of the helium gas molecules is $6.21 \times 10^{-21} \text{ J}$.

- (b) (i) At the same temperature, one mole of helium gas molecules has the same average kinetic energy as one mole of neon gas molecules. The ratio of the average kinetic energy of helium gas molecules to that of neon gas molecules is 1 : 1.

- (ii) Since the molecules of both gases have the same average kinetic energy, i.e.

$$\frac{1}{2}m_{\text{He}}\overline{c_{\text{He}}^2} = \frac{1}{2}m_{\text{Ne}}\overline{c_{\text{Ne}}^2}$$

As $v_{\text{rms}}^2 = \overline{c^2}$, we have

$$\frac{v_{\text{rms(He)}}}{v_{\text{rms(Ne)}}} = \sqrt{\frac{\overline{c_{\text{He}}^2}}{\overline{c_{\text{Ne}}^2}}}$$

$$= \sqrt{\frac{m_{\text{Ne}}}{m_{\text{He}}}}$$

$$= \sqrt{\frac{20}{4}} = \frac{\sqrt{5}}{1}$$

The ratio of the root-mean-square speed of helium gas molecules to that of neon gas molecules is $\sqrt{5} : 1$.

- (c) As the helium behaves as an ideal gas, the potential energy of the gas molecules is zero. Therefore, the change in internal energy of the gas is equal to the change in the total kinetic energy of the gas molecules.

energy required = number of molecules $\times$ increase in average kinetic energy

$$= N_A \times \left(\frac{3RT_f}{2N_A} - \frac{3RT_i}{2N_A} \right)$$

$$= \frac{3R}{2}(T_f - T_i)$$

$$= \frac{3 \times 8.31 \times 50}{2}$$

$$= 623.25 \text{ J}$$

The energy required is 623.25 J.